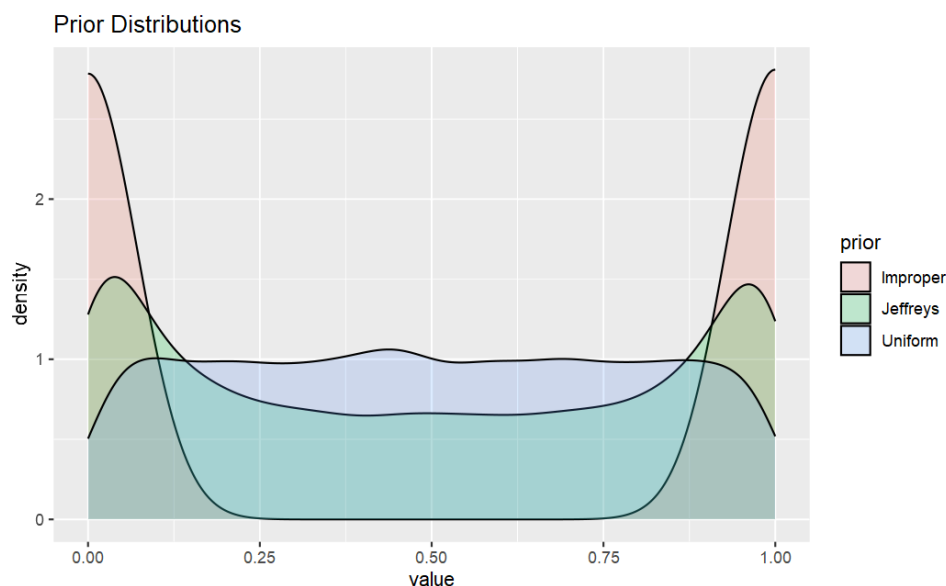


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*** all R code provided at end of document

Problem 1

a)



The above plot overlays the three separate choices for prior distributions - Uniform being Beta(1,1), Jeffreys as Beta(0.5,0.5), and the Improper as Beta(0,0).

The curved densities near the bounds (namely for the improper and for the uniform prior) are likely due to asymptotics where points are rarely generated at the far extreme values. The curvature for the improper prior is probably occurring for readability despite it only generating values of 0 or 1, nothing between

b)

Exam 2 - Alex Caramanico

Problem 1b)

$$Y|X \sim \text{Bin}(N, \theta), \theta \sim \text{Beta}(\alpha, \beta) \quad Y = 241945, N = 493472$$

$$\theta|Y \propto P(Y|\theta) \cdot \pi(\theta)$$

$$\propto \theta^Y (1-\theta)^{n-Y} \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1} = \theta^{\alpha+Y-1} (1-\theta)^{\beta+n-Y-1}$$

new alpha: $\alpha+Y$, new beta: $\beta+n-Y$

$\Rightarrow$ Posterior: $\text{Beta}(\alpha+Y, \beta+n-Y)$

i. $\alpha = \beta = 1$

$\Rightarrow$ Posterior: $\text{Beta}(241946, 251528)$

ii. $\alpha = \beta = 0.5$

$\Rightarrow$ Posterior: $\text{Beta}(241945.5, 251527.5)$

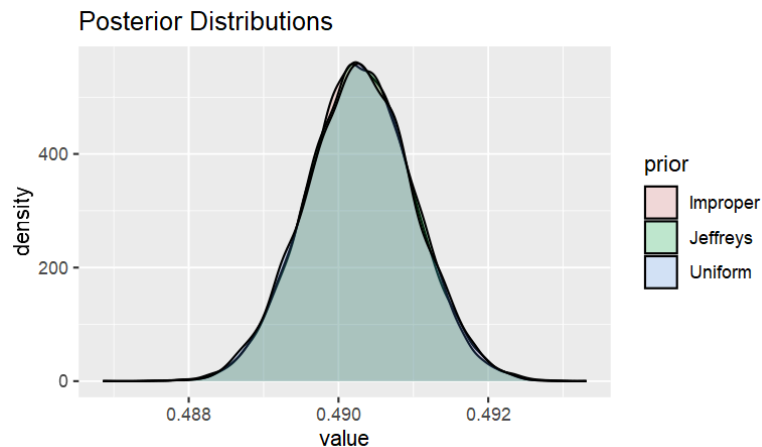
iii. $\alpha = \beta = 0$

$\Rightarrow$ Posterior: $\text{Beta}(241945, 251527)$

Normalizing constant

$$\frac{\binom{n}{Y}}{B(\alpha+\beta)} \cdot \int_0^1 P(Y|\theta) \pi(\theta) d\theta$$

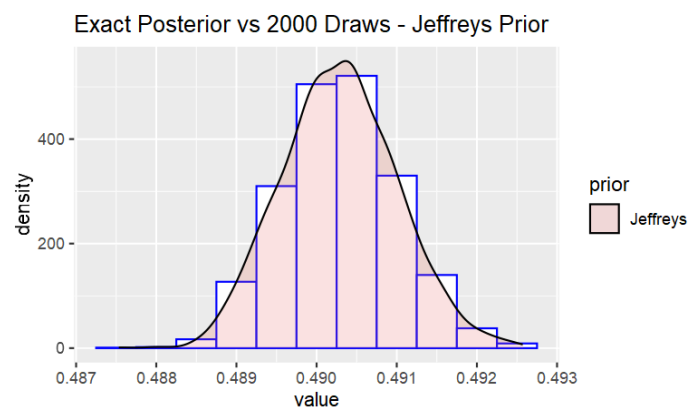
c)



The posterior distributions all appear to be extremely similar due to the large dataset. Beta parameters changing by a value of 1 when they are in the hundreds of thousands will not have a significant effect on the shape of the distribution.

d)

The results of 2000 independent draws from the Beta posterior using Jeffreys prior. It seems clear that the data is dominating the prior due to its magnitude, so the posterior is heavily concentrated around 0.49 which results from Y/N where Y is given to be 241945, N is 493472.



e)

The results for i. and ii. were calculated by taking the mean and variance of the generated values, while the results for iii. and iv. were calculated using `pbeta()` and `qbeta()` respectively, since the posterior is a Beta distribution with known parameters. The probability question would be zero if `pbeta()` was not used as the maximum value of the 2000 draws was 0.4925299.

i. $E(\theta|Y = 241945) = 0.490303$

ii. $Var(\theta|Y = 241945) = 4.929266 \cdot 10^{-7}$

iii. $P(\theta > 0.5|Y = 241945) = 1.145841 \cdot 10^{-42}$

iv. 95% credible interval for $\theta = (0.4888965, 0.4916861)$

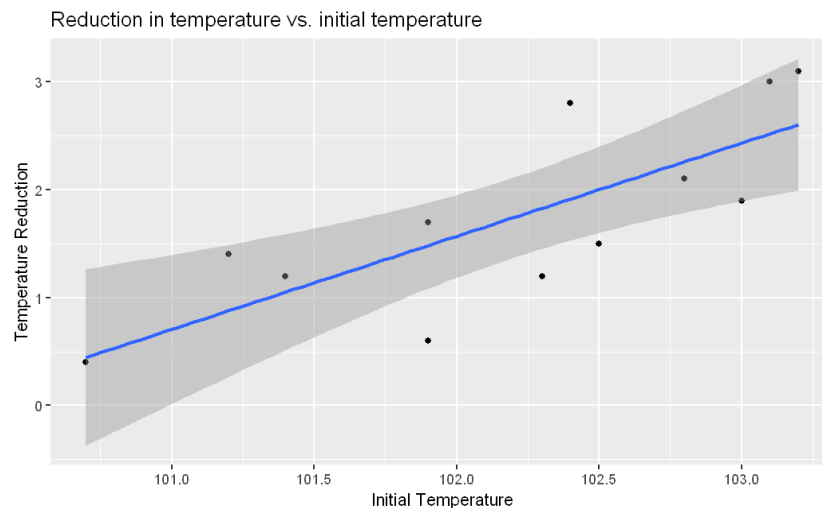
f)

The following results were calculated from the `stanfit` model using the `extract()` function and various in-built R functions. The posterior variance for this approach and for e) are extremely low because of the data dominating the prior. The same can be said for the extremely low probability that θ exceeds 0.5 based on the 2000 samples.

Mean	Variance	P	CredLow	CredUp
0.4902691	5.085591e-07	0	0.4888463	0.4916562

Problem 2

a)



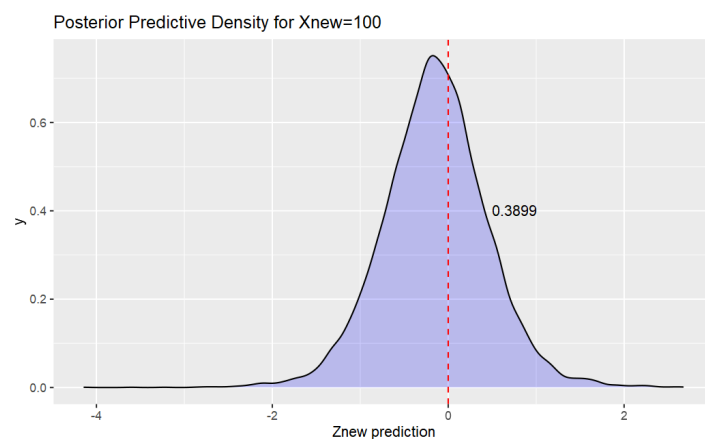
A linear model does appear plausible, as there appears to be a positive relationship between initial temperature and temperature reduction. The line of best fit suggests the same, with its variance bands still representing a linear relationship between the variables.

b)

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta0	1.74	0.00	0.19	1.36	1.63	1.74	1.87	2.13	7562	1
beta1	0.86	0.00	0.26	0.35	0.70	0.86	1.02	1.37	7205	1
tau	0.66	0.00	0.17	0.42	0.54	0.63	0.74	1.09	5095	1

The 95% credible interval for initial temperature effect: (0.35,1.35), results that match the intuition of the researchers. This indicates that higher initial values lead to a greater reduction in temperature. Using RJAGS, the DIC was calculated to be 26.149 (approximately 26, exact values vary on model fit), along with effective model size of 3.948478 (approximately 4). These values will be more useful when compared to the model created in a later part.

c)



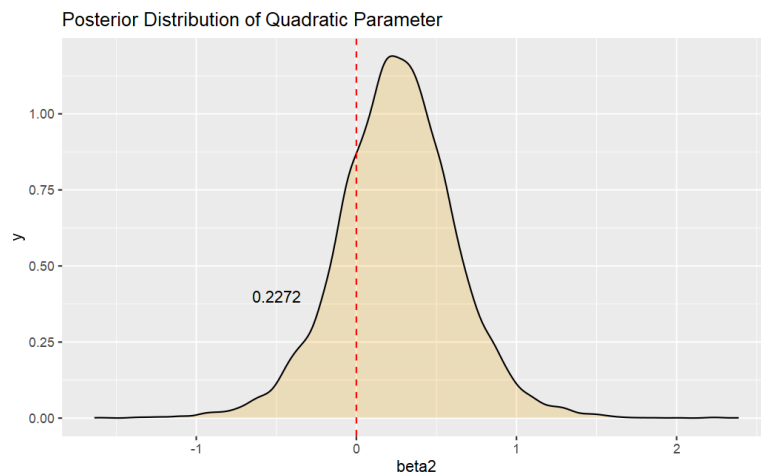
The plot to the left shows the predictive density for Z13, the area to the right of the red line indicating the probability the new patient's temperature is reduced by aspirin given their initial temperature was 100. This probability is shown in the plot as well, and was calculated to be 0.3899.

d)

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
beta0	1.59	0.00	0.30	1.00	1.41	1.59	1.78	2.19	4215	1
beta1	0.95	0.00	0.30	0.35	0.77	0.95	1.13	1.53	5103	1
beta2	0.25	0.01	0.37	-0.48	0.03	0.25	0.48	0.98	4080	1
tau	0.69	0.00	0.20	0.42	0.55	0.65	0.78	1.18	3700	1

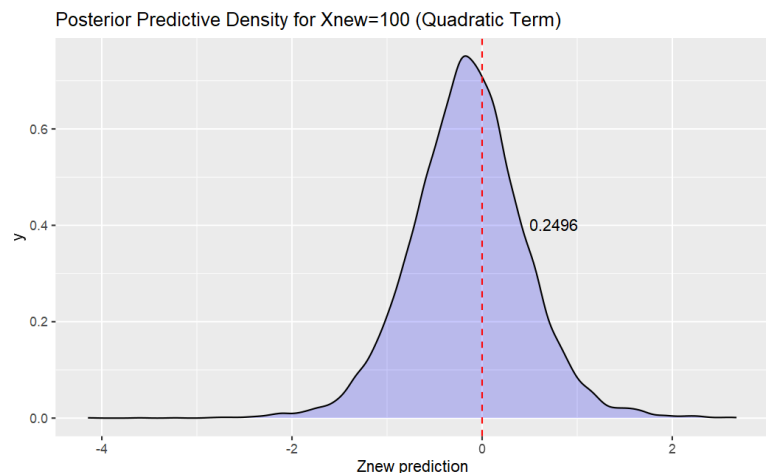
The quadratic term does not appear to be necessary, the 95% credible interval includes zero suggesting it does not have a significant effect on the reduction in temperature. Looking at the data, there does not appear to be much of a curvilinear trend, which supports this result.

For this model, DIC was 29.09892 with effective model size of 6.324443. Both of these values are larger than the first model, suggesting the first model is more useful and the second is unnecessarily complex.



The posterior density of the quadratic parameter also demonstrates that it is not necessary to improve the model, as a considerable portion of it is concentrated to the left of 0. Specifically, more than 22% of its draws were less than 0, suggesting there is not a strong indication of it being positive or negative, thus it can be deemed insignificant.

The predictive density of the new change in temperature is shown to the right. We see that under the new model, the probability that the new patient's temperature is reduced by aspirin given their initial temperature was 100 is actually lower at 0.2496.



Problem 3

$$p(\theta|y) = \frac{\prod_{i=1}^5 [1 + (\theta - y_i)^2]^{-1} I(\theta > 0)}{\int_0^{\infty} \prod_{i=1}^5 [1 + (\theta - y_i)^2]^{-1} d\theta}.$$

The problem is to find the posterior mean and variance of the above pdf using several methods. Results from implementing the five methods are displayed below. The numerical integration approach utilized the `adaptIntegrate()` function from the `cubature` library. The proposal distribution for importance sampling and rejection sampling was normal with mean and variance calculated directly from the observed data vector $y = (4.0, 5.5, 7.5, 3.5, 3)'$, giving $\bar{y} = 4.9$ and $Var(y) = 2.925$ for a proposal distribution of $N(4.9, 2.925)$. All code is included in a separate file.

Approach	PosteriorMean	PosteriorVariance
Numerical Integration	4.556619	0.5577738
Monte Carlo	4.523953	0.6114599
Importance Sampling	4.561365	0.5669065
Rejection Sampling	4.578790	0.9763728
MCMC	4.546359	0.5451127

```

library(rstan)
library(ggplot2)
library(cubature)
library(R2jags)
set.seed(707)
# Problem 5
# a
prior1 <- data.frame(value = rbeta(10000,1,1)) # create data frames for each prior
prior2 <- data.frame(value = rbeta(10000,0.5,0.5))
prior3 <- data.frame(value = rbeta(10000,0,0))
prior1$prior <- 'Uniform' # label generated values
prior2$prior <- 'Jeffreys'
prior3$prior <- 'Improper'
prior_dists <- rbind(prior1,prior2,prior3) # combine into one data frame for plotting
ggplot(data=prior_dists, aes(x=value, fill = prior)) +
  geom_density(alpha = 0.2) +
  ggtitle('Prior Distributions')
# b and c
# Binomial Posterior with Beta prior is Beta(alpha + y, beta + n - y)
N <- 493472 # data values
Y <- 241945
post1 <- data.frame(value = rbeta(10000,1+Y,1+N-Y)) # same process as before but with
posteriors
post2 <- data.frame(value = rbeta(10000,0.5+Y,0.5+N-Y))
post3 <- data.frame(value = rbeta(10000,0+Y,0+N-Y))
post1$prior <- 'Uniform'
post2$prior <- 'Jeffreys'
post3$prior <- 'Improper'
posterior_dists <- rbind(post1,post2,post3)
ggplot(posterior_dists, aes(value, fill = prior)) +
  geom_density(alpha = 0.2) +
  ggtitle('Posterior Distributions')
# d
post_jeffrey <- data.frame(value=rbeta(2000,0.5+Y,0.5+N-Y)) # 2000 draws from posterior
post_jeffrey$prior <- 'Jeffreys'
ggplot(data=post_jeffrey)+
  geom_histogram(aes(x=value),binwidth=0.0005,color='blue',fill='white') +
  geom_density(aes(x=value,fill=prior),alpha=0.2) +
  ggtitle('Exact Posterior vs 2000 Draws - Jeffreys Prior')

```

```

# e
max(post_jeffrey$value) # checking if any were generated >0.5
mean(post_jeffrey$value) # i.  $E[\theta|y=241945]$ 
var(post_jeffrey$value) # ii.  $V[\theta|y=241945]$ 
pbeta(0.5,0.5+Y,0.5+N-Y,lower.tail=FALSE) # iii.  $P(\theta>0.5|y=241945)$ 
jeff_cred <- qbeta(c(0.975,.025),0.5+Y,0.5+N-Y,
                  lower.tail=FALSE) # iv. 95% credible interval
jeff_cred
# f
p5code <- 'data {int<lower = 0> y;
            int<lower = 0> n;}
parameters {real theta;}
model {y~binomial(n,theta);
        theta~beta(0.5,0.5);}'
p5.model <- stan_model(model_code=p5code,model_name='Jeffrey Beta')
p5.draws <- sampling(p5.model,data = list(y = Y,n=N),
                    chains = 2,cores = 2,iter = 10000,warmup = 5000,seed = 500)
p5results <- extract(p5.draws) # extract results to make inferences
pfem <- p5results$theta
credint <- quantile(pfem,c(0.025,0.975))
cbind(Mean=mean(pfem),Variance=var(pfem),
      P=sum(pfem>0.5)/length(pfem),CredLow=credint[1],CredUp=credint[2])

```

```

# Problem 2
patient <- 1:12
x6 <- c(102.4,103.2,101.9,103,101.2,100.7,
        102.5,103.1,102.8,102.3,101.9,101.4) # initial values
y6 <- c(99.6,100.1,100.2,101.1,99.8,100.3,
        101,100.1,100.7,101.1,101.3,100.2) # after values
p6data <- as.data.frame(cbind(patient,x6,y6))
# a
p6data$z6 <- x6-y6 # difference, or reduction
ggplot(data=p6data,mapping=aes(x=x6,y=z6))+
  geom_point()+
  geom_smooth(method='lm')+
  xlab('Initial Temperature')+
  ylab('Temperature Reduction') +
  ggtitle('Reduction in temperature vs. initial temperature')

```

```

# b
scaledx <- p6data$x6-mean(p6data$x6) # so input is in same range as output
slr.stan.code <- '
data{
  int<lower = 0> n;
  array[n] real x;
  array[n] real z;}
parameters{
  real beta0;
  real beta1;
  real<lower=0> tau;}
model{
  for (i in 1:n){
    z[i] ~ normal(beta0 + beta1*x[i],tau);}
  }'

# betas generated from non-informative priors since priors not specified in code
ex6.model = stan_model(model_code=slr.stan.code,model_name='ex6')
ex6.draws = sampling(ex6.model,data = list(n=12,x=scaledx,z=p6data$z6),
  chains = 2,cores = 2,iter = 10000,warmup = 5000,seed = 303)
print(ex6.draws)
slr.jags <- '
model {
  for (i in 1:n) {
    z[i] ~ dnorm(beta0 + beta1 * x[i], tau)}
  beta0 ~ dnorm(0, 0.001)
  beta1 ~ dnorm(0, 0.001)
  tau ~ dgamma(0.1, 0.1)}'
ex6.jagsmodel <- jags(textConnection(slr.jags),data = list(n=12,x=scaledx,z=p6data$z6),
  parameters.to.save = c('beta0', 'beta1', 'tau'),
  inits=NULL,n.chains=2,n.iter=10000,n.burnin=5000,
  jags.seed=303,DIC=TRUE)
ex6.jagsmodel$BUGSoutput$pD
ex6.jagsmodel$BUGSoutput$DIC

# c
ex6_predict <- '
data{
  int<lower=0> n;
  real xNew;
  array[n] real x;

```



```

    array[n] real z;}
parameters{
  real beta0;
  real beta1;
  real<lower=0> tau;}
model{
  for (i in 1:n){
    z[i] ~ normal(beta0 + beta1*x[i],tau);}
  }
generated quantities{
  real z13;
  z13 <- beta0 + beta1*(xNew);}
,

# betas generated from non-informative priors since priors not specified in code
ex6pred.model <- stan_model(model_code=ex6_predict,model_name='ex6pred')
ex6pred.draws <- sampling(ex6pred.model,
  data = list(n=12,xNew=100-mean(p6data$x6),x=scaledx,z=p6data$z6),
  chains = 2,cores = 2,iter = 10000,warmup = 5000,seed = 303)
print(ex6pred.draws)
prediction.ex <- extract(ex6pred.draws)
newzedz <- prediction.ex$z13
ggplot(data.frame(newzedz), aes(x = newzedz)) +
  geom_density(fill = "blue", color="black", alpha = 0.2) +
  ggtitle('Posterior Predictive Density for Xnew=100') +
  xlab('Znew prediction') +
  geom_vline(xintercept=0,linetype='dashed',color='red') +
  annotate('text',x=0.75,y=0.4,label=sum(newzedz>0)/length(newzedz))
# d
slr.stan.code2 <- '
data{
  int<lower = 0> n;
  array[n] real x;
  array[n] real z;
  real xNew;}
parameters{
  real beta0;
  real beta1;
  real beta2;
  real<lower=0> tau;}
model{

```

```

    for (i in 1:n){
      z[i] ~ normal(beta0 + beta1*x[i] + beta2*x[i]^2,tau);}
    }
  generated quantities{
    real z13;
    z13 <- beta0 + beta1*(xNew);}'
# betas generated from non-informative priors since priors not specified in code
ex6.model2 <- stan_model(model_code=slr.stan.code2,model_name='ex6b')
ex6.draws2 <- sampling(ex6.model2,data = list(n=12,x=scaledx,z=p6data$z6,xNew=-2.2),
                      chains = 2,cores = 2,iter = 10000,warmup = 5000,seed = 303)
print(ex6.draws2)
slr.jags2 <- '
model {
  for (i in 1:n) {
    z[i] ~ dnorm(beta0 + beta1 * x[i] + beta2 * x[i]^2, tau)}
  beta0 ~ dnorm(0, 0.001)
  beta1 ~ dnorm(0, 0.001)
  beta2 ~ dnorm(0, 0.001)
  tau ~ dgamma(0.001, 0.001)
}
'

ex6.jagsmodel2 <- jags(textConnection(slr.jags2),data = list(n=12,x=scaledx,z=p6data$z6),
                      parameters.to.save = c('beta0', 'beta1','beta2', 'tau'),
                      inits=NULL,n.chains=2,n.iter=10000,n.burnin=5000,
                      jags.seed=303,DIC=TRUE)
ex6.jagsmodel2$BUGSoutput$pD
ex6.jagsmodel2$BUGSoutput$DIC

prediction.ex2 <- extract(ex6.draws2)
beta2.ex <- prediction.ex2$beta2
ggplot(data.frame(beta2.ex),aes(x=beta2.ex)) +
  geom_density(fill='orange',color='black',alpha=0.2) +
  ggtitle('Posterior Distribution of Quadratic Parameter') +
  xlab('beta2') +
  geom_vline(xintercept=0,linetype='dashed',color='red') +
  annotate('text',x=-.5,y=0.4,label=sum(beta2.ex<0)/length(beta2.ex))

newzedz2 <- prediction.ex2$z13 # get predictions of new reduction
ggplot(data.frame(newzedz), aes(x = newzedz)) +
  geom_density(fill = "blue", color="black", alpha = 0.2) +

```

```

ggtitle('Posterior Predictive Density for Xnew=100 (Quadratic Term)') +
xlab('Znew prediction') +
geom_vline(xintercept=0,linetype='dashed',color='red') +
annotate('text',x=0.75,y=0.4,label=sum(newzedz2>0)/length(newzedz2))

```

Problem 7

```

y7 <- c(4,5.5,7.5,4.5,3) # set up data and posterior
posterior7 <- function(theta,y) {prod(1/(1+(theta-y)^2))*ifelse(theta>0,1,0)}

```

a

```

integral <- adaptIntegrate(function(theta) posterior7(theta, y7), # cubature numerical integration
function

```

```

                lowerLimit = 0,upperLimit = Inf)
normconst <- integral$integral

```

```

post_normed <- function(theta, y) {posterior7(theta, y7) / normconst}

```

```

postmeanNI <- adaptIntegrate(function(theta)
theta * post_normed(theta, y7), lowerLimit = 0, upperLimit = Inf)$integral

```

```

postvarNI <- adaptIntegrate(function(theta)
(theta - postmeanNI)^2 * post_normed(theta, y7),
lowerLimit = 0,upperLimit = Inf)$integral

```

```

NIresults<- data.frame(Approach='Numerical Integration',
PosteriorMean=postmeanNI,PosteriorVariance=postvarNI)

```

b

```

thetas <- runif(10000, 0, 100)
post_samples <- sapply(thetas, function(theta) posterior7(theta, y7)) # posterior for each sample
post_sampNormed <- post_samples / sum(post_samples) # normalize
postmeanMC <- sum(thetas * post_sampNormed)
postvarMC <- sum((thetas - postmeanMC)^2 * post_sampNormed)
MCresults<- data.frame(Approach='Monte Carlo',
PosteriorMean=postmeanMC,PosteriorVariance=postvarMC)

```

c

```

ISthetas <- rnorm(10000,mean(y7),sd(y7))
imp_weights <- sapply(ISthetas, function(theta) posterior7(theta,y7) /
dnorm(theta,mean(y7),sd(y7))) # IS algorithm
imp_weights_normalize <- imp_weights / sum(imp_weights) # normalize
postmeanIS <- sum(ISthetas*imp_weights_normalize) # summary statistics
postvarIS <- sum((ISthetas-postmeanIS)^2*imp_weights_normalize)

```

```

ISresults<- data.frame(Approach='Importance Sampling',
                        PosteriorMean=postmeanIS,PosteriorVariance=postvarIS)
# d
RSttheta <- rnorm(10000,mean(y7),sd(y7))
Paccept <- sapply(RSttheta, function(theta) posterior7(theta,y7) /
                  (50*dnorm(theta,mean(y7),sd(y7)))) # RS algorithm
acc_samps <- RSttheta[runif(10000)<Paccept]
RSresults<- data.frame(Approach='Rejection Sampling',
                        PosteriorMean=mean(acc_samps),PosteriorVariance=var(acc_samps))
# e
m7code <- '
data{
  int<lower=0> n;
  array[n] real y;}
parameters{
  real theta;}
model{
  for (i in 1:n){
    y[i]~cauchy(theta,1);}
  }'
# theta will be generated from flat prior by not specifying in stan code
p7model <- stan_model(model_code=m7code,model_name='Problem7 MCMC')
p7draws <- sampling(p7model, data=list(y=y7,n=5),
                   chains=2,cores=2,iter=10000,warmup=5000,seed=901)
MCMCmean <- get_posterior_mean(p7draws)[1,3] # access mean of two chains
MCMCvar <- var(extract(p7draws)$theta) # calculate variance of chains
MCMCresults<- data.frame(Approach='MCMC',
                          PosteriorMean=MCMCmean,PosteriorVariance=MCMCvar)
rbind(NIresults,MCresults,ISresults,RSresults,MCMCresults) # combine all results

```